

TREE

The tree is mainly used to represent data containing a hierarchical relationship between elements. For example... Records, family trees and table of contents.

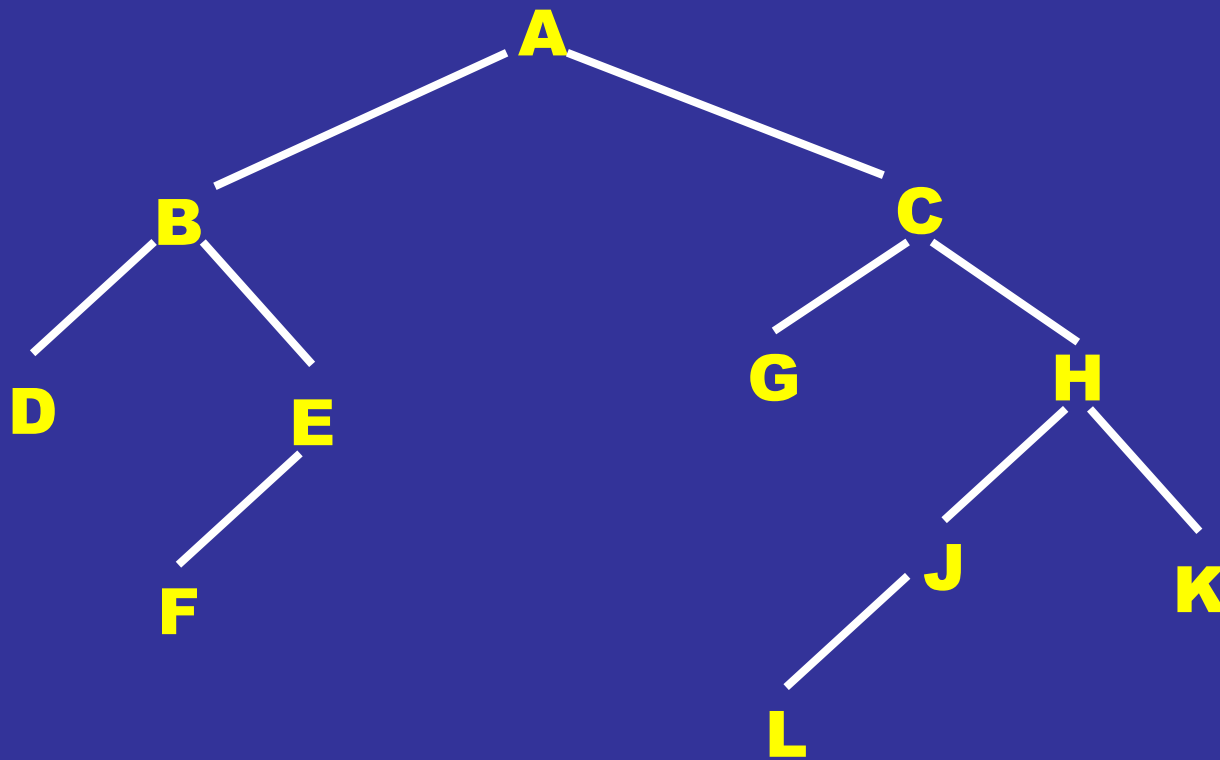
A tree is collection of nodes, that originate from a unique starting node called the **root**.

Nodes are connected by lines called **edges**.

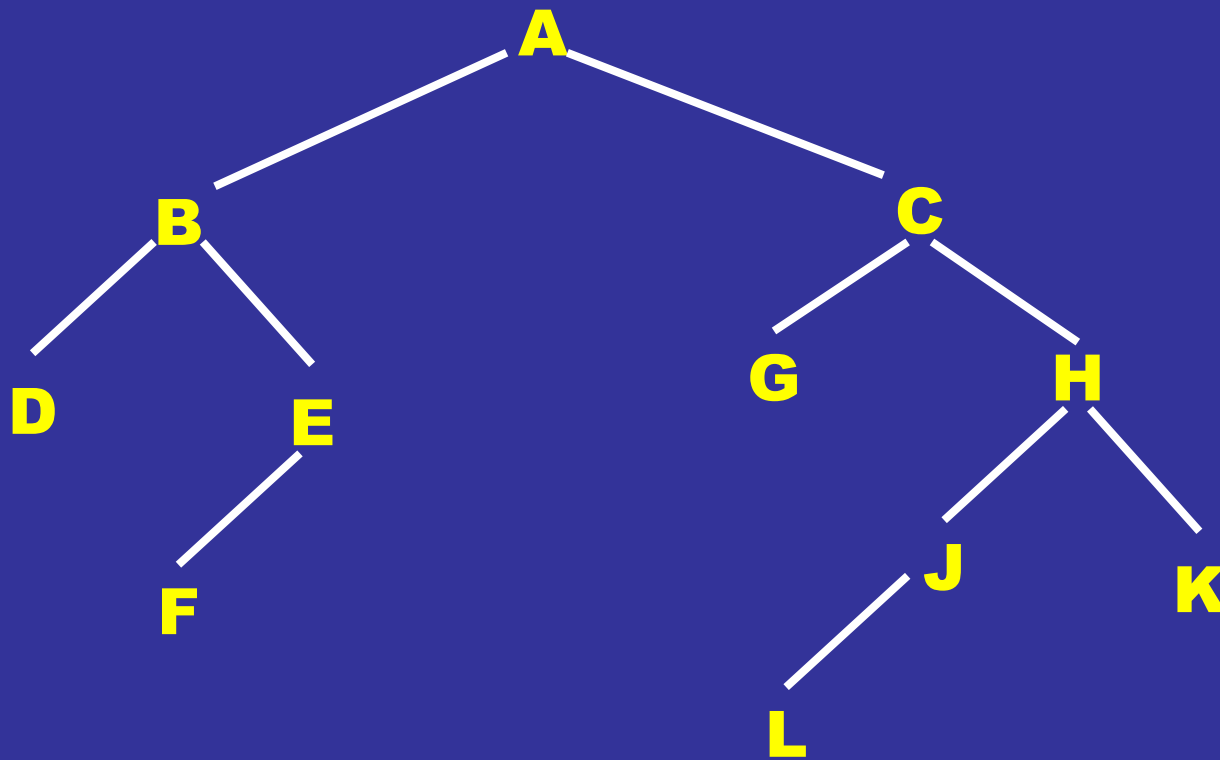
BINARY TREE

A binary tree is defined as a finite set of elements, called nodes, such that

- T is empty (called the Null Tree or Empty Tree)
- T is partitioned into three disjoint subtrees
 - . A single node r , the root
 - . Two possibly empty sets that are binary trees, called left and right subtrees



1. This TREE consists of 11 nodes represented by the letters A through L.
2. The root of TREE is the node A at the top of the diagram.
3. B is the left successor of the node A.
4. C is the right successor of the node A.

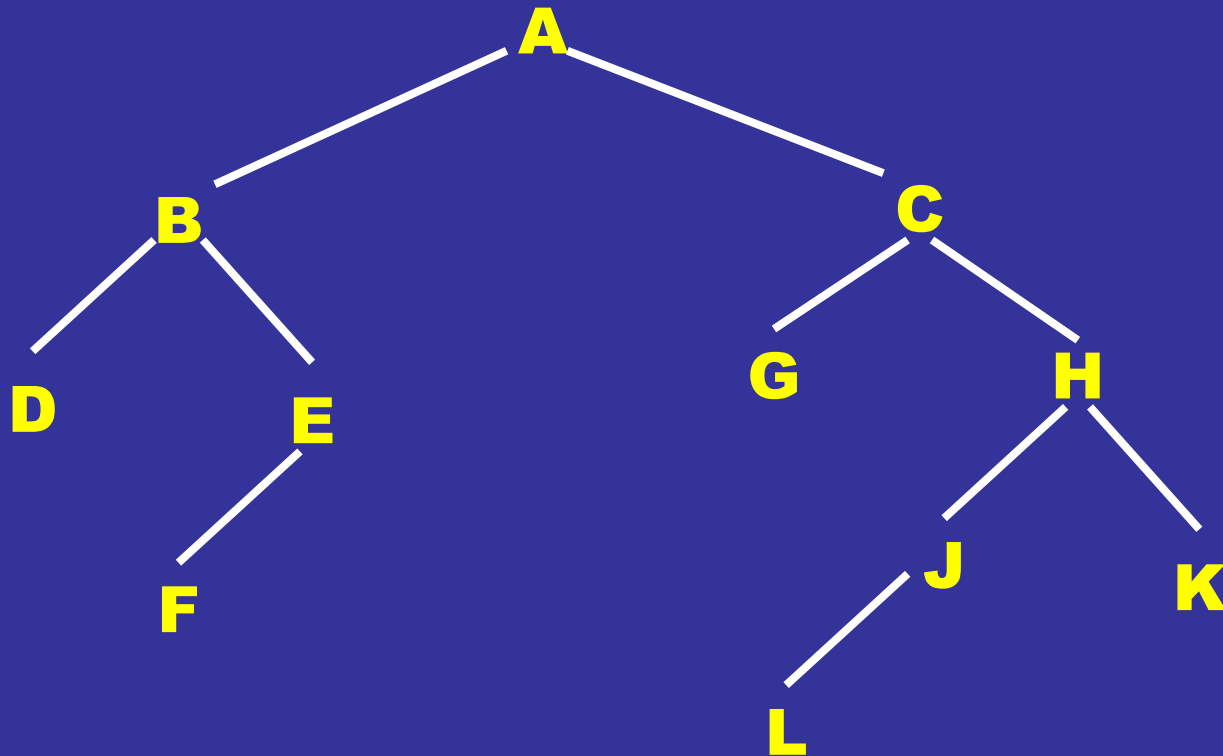


1. The nodes A, B, C and H have two successors.
2. The nodes E and J have only one successor.
3. The nodes D, F, G, L and K have no successor.
4. The nodes with no successor are called **terminal nodes**.

TERMINOLOGY

Suppose N is a node in a tree with left successor S1 and right successor S2

- N is called the PARENT of S1 and S2.
- S1 is called the LEFT CHILD of N.
- S2 is called the RIGHT CHILD of N.
- S1 and S2 are said to be SIBLINGS.
- Nodes are connected by lines called EDGES.
- Sequence of consecutive edges is called a PATH.
- A terminal node is called a LEAF.



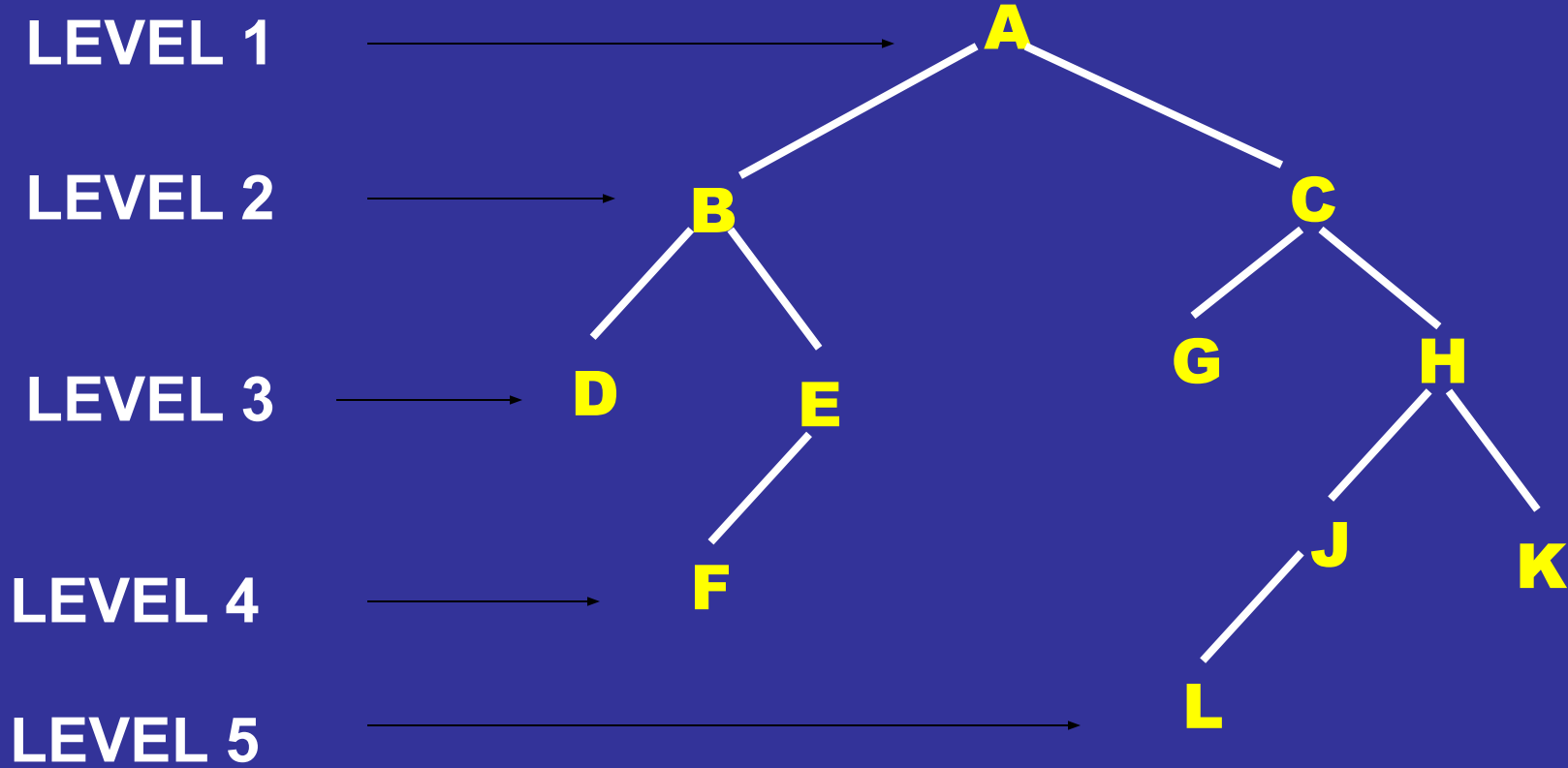
1. B, C, D, E, F, G, H, J, K and L are DESCENDANT of A.
2. H and J are ANCESTORS of L.

HEIGHT OF TREE

The height of any tree is the number of nodes on the longest path from the root to a leaf

The height of a tree T in terms of the levels of its nodes is defined as follows:

- . If T is empty, its height is 0.
- . If T is not empty, its height is equal to the maximum level of its node.



HEIGHT OF TREE = 5

PRACTICE
Figure 7-6
ON PAGE 218

PRACTICE
Figure 7-7
ON PAGE 218

TRAVERSING BINARY TREES

There are three standard ways of traversing a binary tree

PREORDER

- Process the root R
- Traverse the left subtree of R in preorder
- Traverse the right subtree of R in preorder

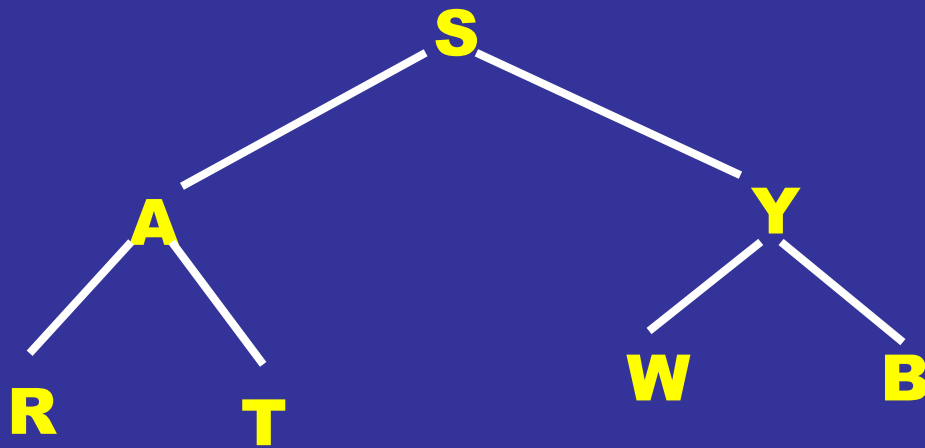
INORDER

- Traverse the left subtree of R in inorder
- Process the root R
- Traverse the right subtree of R in inorder

POSTORDER

- Traverse the left subtree of R in postorder
- Traverse the right subtree of R in postorder
- Process the root R

Representation of Tree as an Array

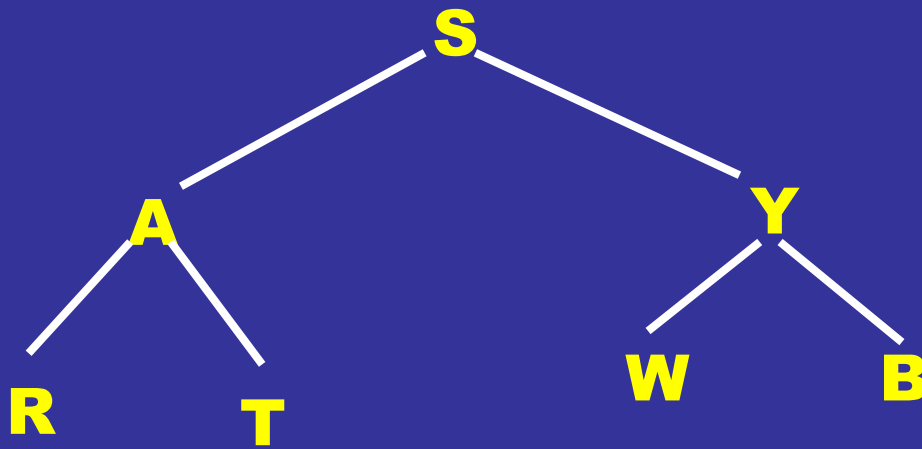


left Child = $2n$
right Child = $2n+1$

n is location of the parent

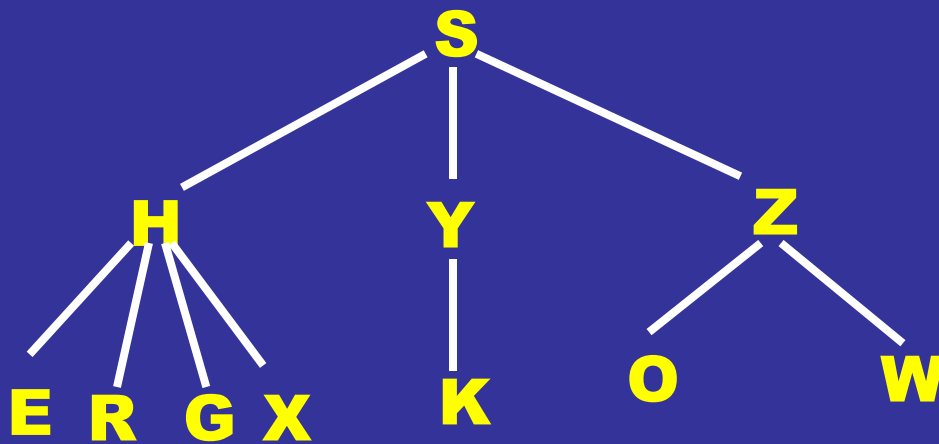
1	S
2	A
3	Y
4	R
5	T
6	W
7	B

Representation of Tree as an Linked List



	info	left	right
1	A	3	7
2	B	0	0
3	R	0	0
4			8
5	S	1	9
6	W	0	0
7	T	0	0
8			10
9	Y	6	2
10			0

GENERAL TREE



	info	child	sib
1	X	0	0
2	H	11	8
3	O	0	9
4	R	0	7
5			
6			
7	G	0	1
8	Y	13	16
9	W	0	0
10	S	2	0
11	E	0	4
12			
13	K	0	0
14			
15			
16	Z	3	0

HEAP SORT

Heap sort works on the principal of Heap Tree.

Types of Heap Trees

1.MAX. HEAP TREE

2.MIN. HEAP TREE

In Max. Heap Tree:

- Root will be the maximum
- Children will be less then their parents

Initial Step:

50	40	10	60	30	70
----	----	----	----	----	----

